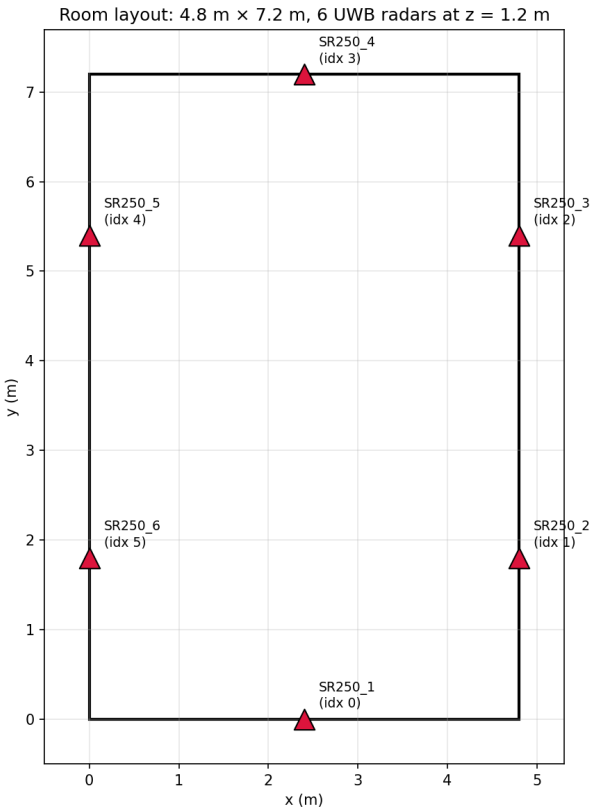


EDA Briefing — Multi-Person UWB Localization

Generated 2026-04-30 11:41 · Dataset: HAEEL/multi-person-localization · 24 windows · 6 radars × 3 antennas × 120 bins @ 25 Hz · Room 4.8×7.2 m · Total 180,000 frames (120.0 min)

1. Sensor rig

Radar idx	Name	x (m)	y (m)	z (m)
0	SR250_1	2.40	0.00	1.20
1	SR250_2	4.80	1.80	1.20
2	SR250_3	4.80	5.40	1.20
3	SR250_4	2.40	7.20	1.20
4	SR250_5	0.00	5.40	1.20
5	SR250_6	0.00	1.80	1.20

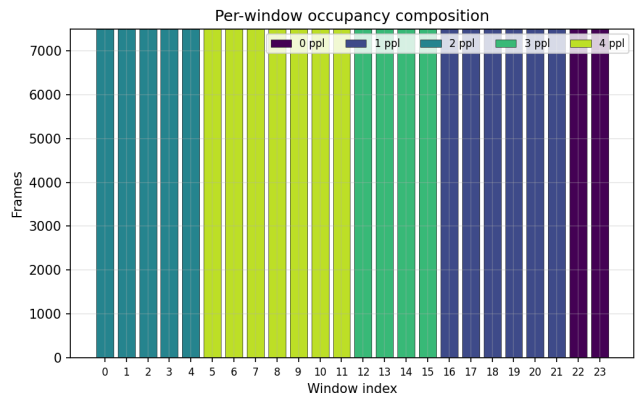
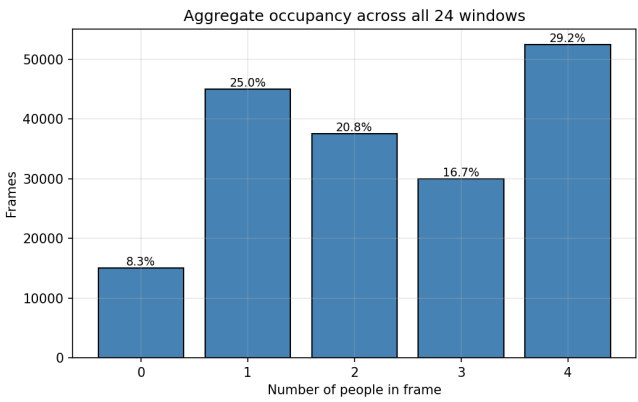


2. Window inventory

Windows	24
Total frames	180,000
Total duration	7199.0 s (120.0 min)
Mean sample rate	25.00 Hz (spec: 25 Hz)
CIR shape (per frame)	(6, 3, 120, 2)
CIR dtype	float32

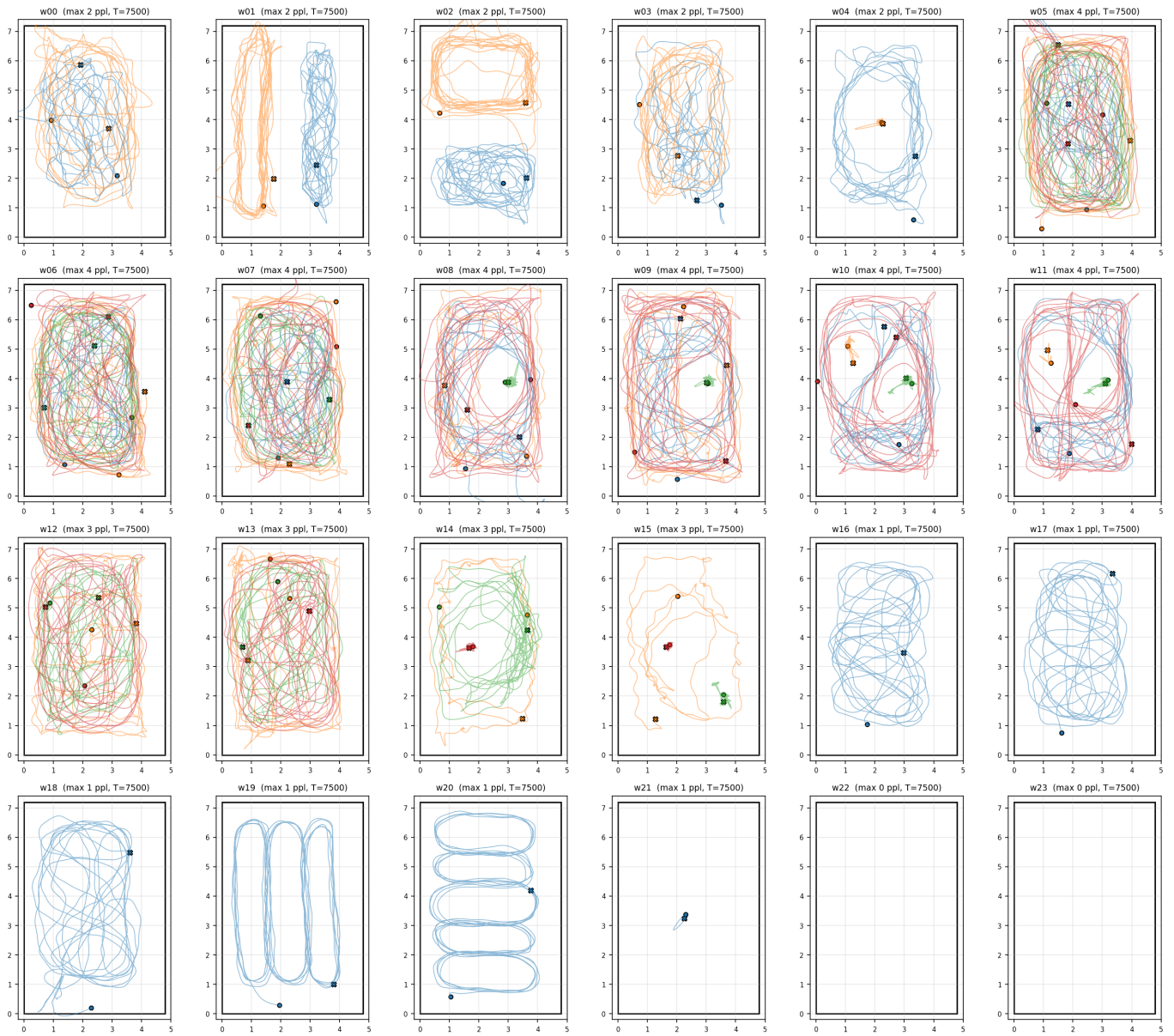
3. Occupancy distribution (aggregate)

# people	Frames	% of dataset
0	15,000	8.33 %
1	45,000	25.00 %
2	37,500	20.83 %
3	30,000	16.67 %
4	52,500	29.17 %



4. Ground-truth trajectories (all 24 windows)

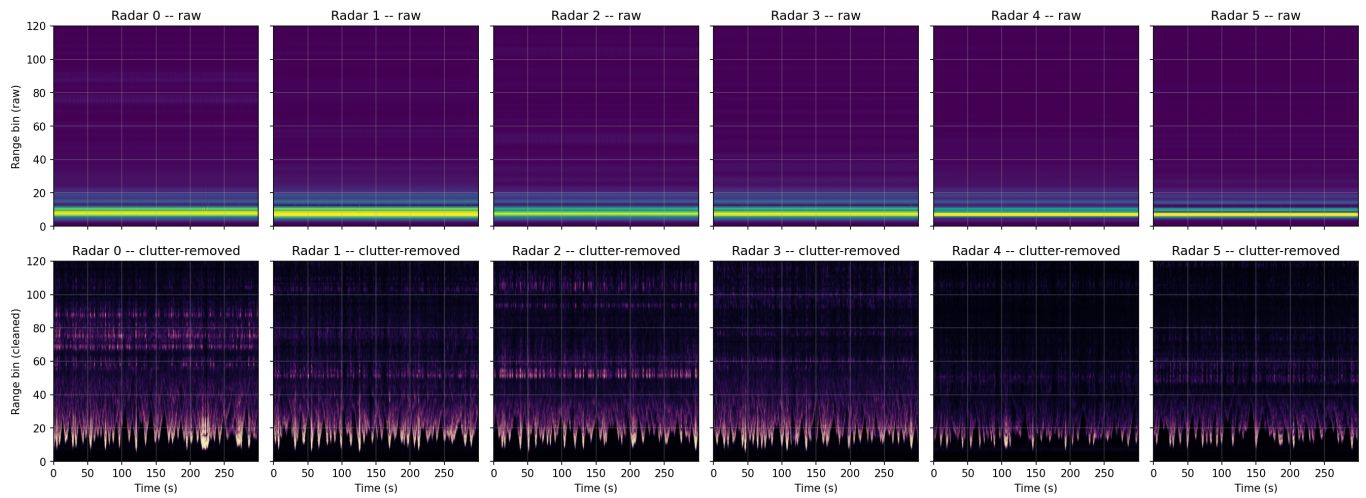
Ground-truth trajectories for all 24 acquisition windows



5. CIR statistics & clutter removal

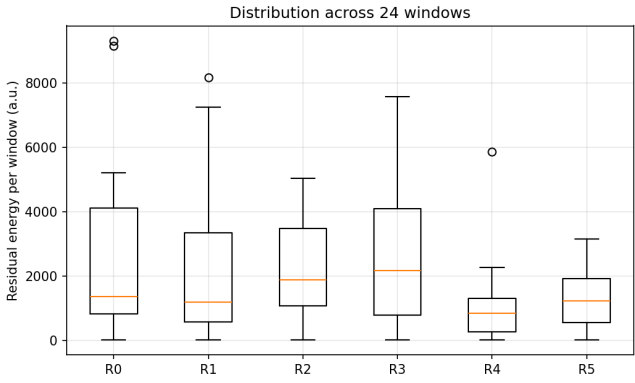
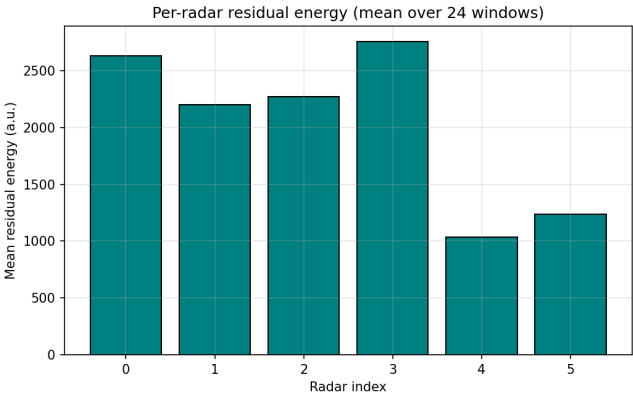
Example window: **window_000005**. |CIR| stats — min=0, max=3.56e+04, mean=1.34e+03, std=3.62e+03, p99=2.1e+04. Frame interval — mean=40.00 ms (std=0.01 ms). Clutter removal: centred running-mean subtraction with $W=50$ frames (≈ 2 s).

Effect of running-mean clutter removal ($W=50$ frames = 2.0 s)



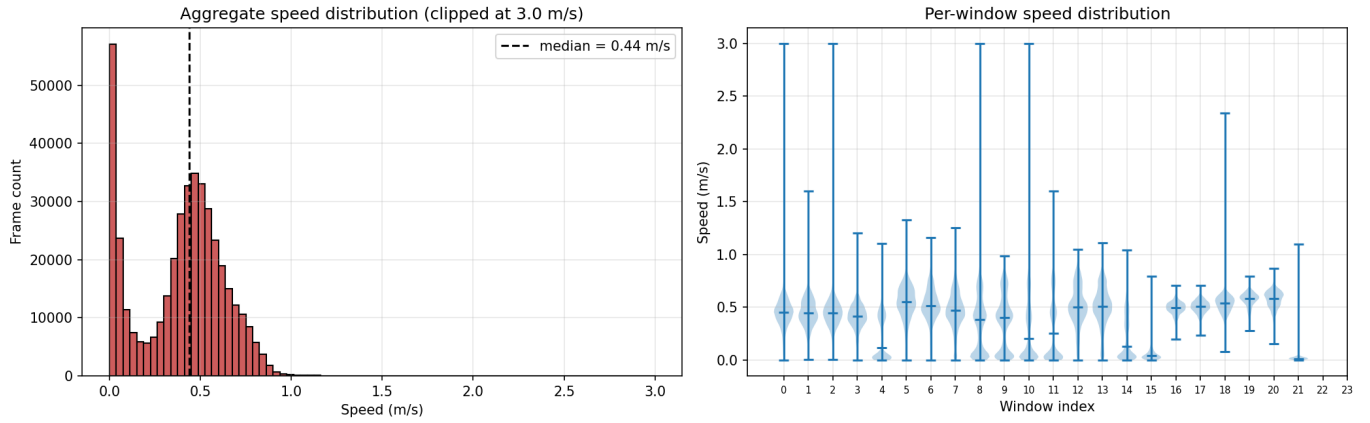
6. Per-radar residual energy (post clutter removal)

Radar	Mean	Median	Std (across windows)
0 (SR250_1)	2633	1355	2580
1 (SR250_2)	2198	1191	2199
2 (SR250_3)	2270	1872	1534
3 (SR250_4)	2756	2180	2256
4 (SR250_5)	1034	839	1192
5 (SR250_6)	1234	1218	826.6



7. Motion / speed statistics (from people_xy)

Stat	Value (m/s)
median	0.438
mean	0.394
p90	0.683
p99	0.858
max	4.005
n samples	419,944



8. Per-window summary

Columns: **w**=window idx, **T**=frames, **dur**=duration (s), **max/mean**=people count, **0p..4p**=frames at each occupancy level, **v_med/v_p90**=speed median / 90th-pct (m/s).

w	T	dur	max	mean	0p	1p	2p	3p	4p	v_med	v_p90
0	7500	300	2	2.00	0	0	7500	0	0	0.45	0.61
1	7500	300	2	2.00	0	0	7500	0	0	0.45	0.62
2	7500	300	2	2.00	0	0	7500	0	0	0.45	0.60
3	7500	300	2	2.00	0	0	7500	0	0	0.41	0.55
4	7500	300	2	2.00	0	0	7500	0	0	0.12	0.48
5	7500	300	4	4.00	0	0	0	0	7500	0.55	0.73
6	7500	300	4	4.00	0	0	0	0	7500	0.52	0.73
7	7500	300	4	4.00	0	0	0	0	7500	0.47	0.70
8	7500	300	4	4.00	0	0	0	0	7500	0.38	0.71
9	7500	300	4	4.00	0	0	0	0	7500	0.40	0.73
10	7500	300	4	4.00	0	0	0	0	7500	0.20	0.71
11	7500	300	4	4.00	0	0	0	0	7500	0.25	0.75
12	7500	300	3	3.00	0	0	0	7500	0	0.51	0.77
13	7500	300	3	3.00	0	0	0	7500	0	0.51	0.75
14	7500	300	3	3.00	0	0	0	7500	0	0.13	0.52
15	7500	300	3	3.00	0	0	0	7500	0	0.04	0.33
16	7500	300	1	1.00	0	7500	0	0	0	0.50	0.56
17	7500	300	1	1.00	0	7500	0	0	0	0.51	0.58
18	7500	300	1	1.00	0	7500	0	0	0	0.54	0.62
19	7500	300	1	1.00	0	7500	0	0	0	0.58	0.65
20	7500	300	1	1.00	0	7500	0	0	0	0.58	0.67
21	7500	300	1	1.00	0	7500	0	0	0	0.01	0.03
22	7500	300	0	0.00	7500	0	0	0	0	0.00	0.00
23	7500	300	0	0.00	7500	0	0	0	0	0.00	0.00

9. metadata.jsonl (raw scenario labels)

window_id	path	num_frames	duration_sec	description	acquisition_timestamp
window_000000	data/window_000000.npz	7500	300.0	random walk of 2 subjects	20260220-112438
window_000001	data/window_000001.npz	7500	300.0	random walk of 2 subjects, one to t	20260220-113155
window_000002	data/window_000002.npz	7500	300.0	random walk of 2 subjects, one abov	20260220-113848
window_000003	data/window_000003.npz	7500	300.0	random walk of 2 subjects	20260220-114524
window_000004	data/window_000004.npz	7500	300.0	random walk of 1 subject while the	20260220-115314
window_000005	data/window_000005.npz	7500	300.0	random walk of 4 subjects	20260306-103626
window_000006	data/window_000006.npz	7500	300.0	random walk of 4 subjects	20260306-104232
window_000007	data/window_000007.npz	7500	300.0	random walk of 4 subjects	20260306-104815
window_000008	data/window_000008.npz	7500	300.0	random walk of 3 subjects and 1 sea	20260306-105449
window_000009	data/window_000009.npz	7500	300.0	random walk of 3 subjects and 1 sea	20260306-110056
window_000010	data/window_000010.npz	7500	300.0	random walk of 2 subjects and 2 sea	20260306-110713
window_000011	data/window_000011.npz	7500	300.0	random walk of 2 subjects and 2 sea	20260306-111311
window_000012	data/window_000012.npz	7500	300.0	random walk of 3 subjects	20260306-112045
window_000013	data/window_000013.npz	7500	300.0	random walk of 3 subjects	20260306-112631
window_000014	data/window_000014.npz	7500	300.0	random walk of 2 subjects and 1 sea	20260306-113219
window_000015	data/window_000015.npz	7500	300.0	random walk of 1 subject and 2 seat	20260306-113809
window_000016	data/window_000016.npz	7500	300.0	random walk of 1 subject	20260306-133216
window_000017	data/window_000017.npz	7500	300.0	random walk of 1 subject	20260306-133810
window_000018	data/window_000018.npz	7500	300.0	random walk of 1 subject	20260306-134348
window_000019	data/window_000019.npz	7500	300.0	serpentine walk of 1 subject follow	20260306-135017
window_000020	data/window_000020.npz	7500	300.0	serpentine walk of 1 subject follow	20260306-140154
window_000021	data/window_000021.npz	7500	300.0	1 subject seated	20260306-140821
window_000022	data/window_000022.npz	7500	300.0	acquisition with 0 subjects	20260306-141428
window_000023	data/window_000023.npz	7500	300.0	acquisition with 0 subjects	20260306-142048